LN5 Stochastic Calculus

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Outlines

Textbook: Stochastic Calculus for Finance II: Continuous-Time Models: Chapter 4

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Itô's Integral for Simple Integrands

We fix a positive number T and seek to make sense of

$$\int_0^T \Delta(t) dW(t).$$

Construction of the integral

Let $\Delta(t)$ be a simple process. For $0 = t_0 \le t_1 \le \cdots \le t_n = T$,

$$\Delta(t) = \sum_{j=o}^{n-1} H_j \mathbf{1}_{[t_j, t_{j+1}]}(t) = \sum_{j=o}^{n-1} \Delta(t_j) \mathbf{1}_{[t_j, t_{j+1})}(t),$$

where H_j are random variables.
For $t_k \le t \le t_{k+1}$, define

$$\begin{aligned} I(t) &= \sum_{j=0}^{k-1} \Delta(t_j)[W(t_{j+1}) - W(t_j)] + \Delta(t_k)[W(t) - W(t_k)] \\ &= \int_0^t \Delta(u) dW(u) \end{aligned}$$

The process $I(t)$ is the Itô's integral of the simple process $\Delta(t)$.

Properties of the Integral

Theorem 1

The Itô's integral defined above is a martingale.

Proof Let $0 \le s \le t \le T$. We assume that s and t are in different sub-intervals. Assume $t_l < t_k, s \in [t_l, t_{l+1}), t \in [t_k, t_{k+1})$.

$$\begin{aligned} I(t) = &\sum_{j=0}^{l-1} \Delta(t_j)(W(t_{j+1}) - W(t_j)) + \Delta(t_l)(W(t_{l+1}) - W(t_l)) \\ &+ \sum_{j=l+1}^{k-1} \Delta(t_j)(W(t_{j+1}) - W(t_j)) + \Delta(t_k)(W(t) - W(t_k)) \end{aligned} \tag{1}$$

Proof Continued

Since $t_l \leq s$,

$$\mathbb{E}\left[\sum_{j=0}^{l-1} \Delta(t_j)(W(t_{j+1}) - W(t_j))|\mathcal{F}(s)\right] = \sum_{j=0}^{l-1} \Delta(t_j)(W(t_{j+1}) - W(t_j)) \tag{2}$$

$$\begin{aligned}\mathbb{E}\left[\Delta(t_l)(W(t_{l+1}) - W(t_l))|\mathcal{F}(s)\right] &= \Delta(t_l)\left(\mathbb{E}[W(t_{j+1})|\mathcal{F}(s)] - W(t_l)\right]) \\ &= \Delta(t_l)\left(W(s) - W(t_l)\right)\end{aligned} \tag{3}$$

Proof Continued

Since $t_j \geq t_{l+1} > s$ in the third term,

$$\begin{aligned}
&\mathbb{E}\left[\Delta(t_j)(W(t_{j+1}) - W(t_j))|\mathcal{F}(s)\right] \\
&= \mathbb{E}\left[\mathbb{E}\left[\Delta(t_j)(W(t_{j+1}) - W(t_j))|\mathcal{F}(t_j)\right]|\mathcal{F}(s)\right] \\
&= \mathbb{E}\left[\Delta(t_j)(W(t_j) - W(t_j))|\mathcal{F}(s)\right] = 0.
\end{aligned} \tag{4}$$

Therefore,

$$\mathbb{E}\left[\sum_{j=l+1}^{k-1} \Delta(t_j)(W(t_{j+1}) - W(t_j))|\mathcal{F}(s)\right] = 0$$

Similarly,

$$\mathbb{E}\left[\Delta(t_k)(W(t) - W(t_k))|\mathcal{F}(s)\right] = 0.$$

Properties of the Integral

Theorem 2

$\mathbb{E}I^2(t) = \mathbb{E}\int_0^t \Delta^2(u)du.$

Proof Set $D_j = W(t_{j+1}) - W(t_j)$ for $j = 0, \cdots, k-1$ and $D_k = W(t) - W(t_k)$.
Then, $I(t) = \sum_{j=0}^{k} \Delta(t_j)D_j$ and

$$I^2(t) = \sum_{j=0}^{k} \Delta^2(t_j)D_j^2 + 2\sum_{0\le i<j\le k} \Delta(t_i)\Delta(t_j)D_iD_j$$

Since $\Delta(t_i)\Delta(t_j)D_i$ is $\mathcal{F}(t_j)$ measurable and D_j is independent of $\mathcal{F}(t_j)$,

$$\mathbb{E}[\Delta(t_i)\Delta(t_j)D_iD_j] = \mathbb{E}[\Delta(t_i)\Delta(t_j)D_i]\mathbb{E}[D_j] = 0$$

Proof Continued

Therefore,

$$\begin{aligned}
\mathbb{E}[I^2(t)] &= \sum_{j=0}^{k} \mathbb{E}[\Delta^2(t_j) D_j^2] \\
&= \sum_{j=0}^{k} \mathbb{E}[\Delta^2(t_j)]\mathbb{E}[D_j^2] \\
&= \sum_{j=0}^{k-1} \mathbb{E}\Delta^2(t_j)(t_{j+1} - t_j) + \mathbb{E}\Delta^2(t_k)(t - t_k) \qquad (5) \\
&= \sum_{j=0}^{k-1} \mathbb{E} \int_{t_j}^{t_{j+1}} \Delta^2(u) du + \mathbb{E} \int_{t_k}^{t} \Delta^2(u) du \\
&= \mathbb{E} \int_{0}^{t} \Delta^2(u) du
\end{aligned}$$

Properties of the Integral

Theorem 3

$[I, I](t) = \int_0^t \Delta^2(u)du$

Proof Let $[t_j, t_{j+1}]$ be a sub-interval. We choose partition points $t_j = s_0 < s_1 < \cdots < s_m = t_{j+1}$ and consider

$$\begin{aligned}\sum_{i=0}^{m-1}[I(s_{i+1}) - I(s_i)]^2 &= \sum_{i=0}^{m-1}[\Delta(t_j)(W(s_{i+1}) - W(s_i))]^2 \\ &= \Delta^2(t_j)\sum_{i=0}^{m-1}(W(s_{i+1}) - W(s_i))^2\end{aligned} \tag{6}$$

Proof Continued

As $m \to \infty$ and the step size approaches zero, the term $\sum_{i=0}^{m-1}(W(s_{i+1}) - W(s_i))^2$ converges to the QV accumulated by Brownian motion between t_j and t_{j+1}, which is $t_{j+1} - t_j$. Therefore,

$$\begin{aligned} \Delta^2(t_j) \sum_{i=0}^{m-1}(W(s_{i+1}) - W(s_i))^2 &\to \Delta^2(t_j)(t_{j+1} - t_j) \\ &= \int_{t_j}^{t_{j+1}} \Delta^2(u)du \end{aligned} \tag{7}$$

Adding up all these pieces, we obtain the theorem. □

Itô's integral for general integrand

In this section, we define the $\int_0^T \Delta(t)dW(t)$ for general integrands $\Delta(t)$. We assume that

(1) $\Delta(t)$ is adapted to the filtration $\mathcal{F}(t)$.

(2) $\mathbb{E}\int_0^T \Delta^2(t)dt < \infty$

We want to approximate $\Delta(t)$ by simple processes. In general, we choose a sequence $\Delta_n(t)$ of simple processes such that as $n \to \infty$ these processes converge to the continuously varying $\Delta(t)$. By "converge", we mean that

$$\lim_{n\to\infty} \mathbb{E}\int_0^T |\Delta_n(t) - \Delta(t)|^2 dt = 0.$$

Then we define the Itô integral for the continuously varying integrand $\Delta(t)$ by the formula

$$\int_0^t \Delta(u)dW(u) = \lim_{n\to\infty} \int_0^t \Delta_n(u)dW(u), 0 \le t \le T.$$

Itô's Integral for General Integrand

Theorem 4

$I(t) = \int_0^t \Delta(u)dW(u)$ *has the following properties.*

(1) *(Continuity)* $I(t)$ *is continuous in* t.

(2) *(Adaptivity)* $I(t)$ *is* $\mathcal{F}(t)$*-measurable for each* t.

(3) *(Linearity) If* $I(t) = \int_0^t \Delta(u)dW(u)$ *and* $J(t) = \int_0^t \Gamma(u)dW(u)$*, then* $I(t) \pm J(t) = \int_0^t (\Delta(u) \pm \Gamma(u))dW(u)$*; furthermore, for every constant* c*,* $cI(t) = \int_0^t c\Delta(u)dW(u)$

(4) *(Martingale)* $I(t)$ *is a martingale.*

(5) *(Itô isometry)* $\mathbb{E}I^2(t) = \mathbb{E}\int_0^t \Delta^2(u)du$.

(6) *(QV)* $[I, I](t) = \int_0^t \Delta^2(u)d(u)$.

Example of Itô's integral for general integrand

We compute $\int_0^T W(t)dW(t)$.

$$\Delta_n(t) = \begin{cases} W(0) = 0, \ \textit{if} \ 0 \le t < \frac{T}{n}, \\ W(\frac{T}{n}), \ \textit{if} \ \frac{T}{n} \le t < \frac{2T}{n}, \\ \vdots \qquad \vdots \\ W(\frac{(n-1)T}{n}), \ \textit{if} \ \frac{(n-1)T}{n} \le t < T \end{cases} \tag{8}$$

Then, $\lim_{n\to\infty} \mathbb{E} \int_0^T |\Delta_n(t) - W(t)|^2 dt = 0$. Let $W_j = W(\frac{jT}{n})$.

$$\begin{aligned} \int_0^T W(t)dW(t) &= \lim_{n\to\infty} \int_0^T \Delta_n(t)dW(t) \\ &= \lim_{n\to\infty} \sum_{j=0}^{n-1} W_j(W_{j+1} - W_j) \quad (*) \end{aligned} \tag{9}$$

Example Continued

$$\begin{aligned}
\frac{1}{2}\sum_{j=0}^{n-1}(W_{j+1}-W_j)^2 &= \frac{1}{2}\sum_{j=0}^{n-1}W_{j+1}^2 - \sum_{j=0}^{n-1}W_jW_{j+1} + \frac{1}{2}\sum_{j=0}^{n-1}W_j^2 \\
&= \frac{1}{2}W_n^2 + \frac{1}{2}\sum_{k=0}^{n-1}W_j^2 - \sum_{j=0}^{n-1}W_jW_{j+1} + \frac{1}{2}\sum_{j=0}^{n-1}W_j^2 \\
&= \frac{1}{2}W_n^2 + \sum_{k=0}^{n-1}W_k^2 - \sum_{j=0}^{n-1}W_jW_{j+1} \\
&= \frac{1}{2}W_n^2 + \sum_{j=0}^{n-1}W_j(W_j - W_{j+1})
\end{aligned} \tag{10}$$

Example Continued

Therefore,

$$
\begin{aligned}
(*) &= \lim_{n\to\infty} \left\{ \frac{1}{2}W^2(T) - \frac{1}{2}\sum_{j=0}^{n-1}[W(\frac{(j+1)T}{n}) - W(\frac{jT}{n})]^2 \right\} \\
&= \frac{1}{2}W^2(T) - \frac{1}{2}[W,W](T) \\
&= \frac{1}{2}W^2(T) - \frac{1}{2}T
\end{aligned}
\tag{11}
$$

Formula for Brownian Motion

Theorem 5

(Itô's formula for Brownian motion) Let $f(t,x) \in C^{1,2}$, and let $W(t)$ be a Brownian motion. Then, for every $T \geq 0$,

$$f(T, W(t)) = f(0, W(0)) + \int_0^T f_t(t, W(t))dt + \int_0^T f_x(t, W(t))dW(t) + \frac{1}{2}\int_0^T f_{xx}(t, W(t))dt. \tag{12}$$

Formula for Brownian Motion

Sketch of proof

(1) Taylor series expansion

(2) higher order terms $\to 0$

(3) Quadratic variation

$$
\begin{aligned}
& f(t_{j+1}, x_{j+1}) - f(t_j, x_j) \\
& = f_t(t_j, x_j)(t_{j+1} - t_j) + f_x(t_j, x_j)(x_{j+1} - x_j) \\
& + \frac{1}{2} f_{xx}(t_j, x_j)(x_{j+1} - x_j)^2 + f_{tx}(t_j, x_j)(t_{j+1} - t_j)(x_{j+1} - x_j) \\
& + \frac{1}{2} f_{tt}(t_j, x_j)(t_{j+1} - t_j)^2 + \text{higher-order terms.}
\end{aligned} \tag{13}
$$

$$
\begin{aligned}
& f(T, W(t)) - f(0, W(0)) \\
&= \sum_{j=0}^{n-1} [f(t_{j+1}, W_{j+1}) - f(t_j, W_j)] \\
&= \sum_{j=0}^{n-1} f_t(t_j, W(t_j))(t_{j+1} - t_j) + \sum_{j=0}^{n-1} f_x(t_j, W(t_j))(W(t_{j+1}) - W(t_j)) \\
&+ \frac{1}{2} \sum_{j=0}^{n-1} f_{xx}(t_j, W(t_j))(W(t_{j+1}) - W(t_j))^2 \\
&+ \sum_{j=0}^{n-1} f_{tx}(t_j, W(t_j))(t_{j+1} - t_j)(W(t_{j+1}) - W(t_j)) \\
&+ \frac{1}{2} \sum_{j=0}^{n-1} f_{tt}(t_j, W(t_j))(t_{j+1} - t_j)^2 + \text{higher-order terms.}
\end{aligned}
\tag{14}
$$

Itô Processes

Definition 1

Let $W(t)$ be a Brownian motion, and let $\mathcal{F}(t)$ be an associated filtration. An Itô process is a stochastic process of the form

$$X(t) = X(0) + \int_0^t \Delta(u)dW(u) + \int_0^t \Theta(u)du,$$

where $X(0)$ is nonrandom and $\Delta(u)$ and $\Theta(u)$ are adapted stochastic processes.

Lemma

The quadratic variation of the Itô process is

$$[X, X](t) = \int_0^t \Delta^2(u)du.$$

Formula for Itô Processes

Definition 2

Let $X(t)$ be an Itô process, and let $\Gamma(t)$ be an adapted process. We define the integral with respect to an Itô process

$$\int_0^t \Gamma(u)dX(u) = \int_0^t \Gamma(u)\Delta(u)dW(u) + \int_0^t \Gamma(u)\Theta(u)du.$$

Itô-Doeblin Formula for an Itô process

Theorem 6

Let $X(t)$ be an Itô process, and let $f(t,x) \in C^{1,2}$. Then, for every $T \geq 0$,

$$\begin{aligned} & f(T, X(T)) \\ &= f(0, X(0)) + \int_0^T f_t(t, X(t))dt + \int_0^T f_x(t, X(t))dX(t) \\ &+ \frac{1}{2}\int_0^T f_{xx}(t, X(t))d[X, X](t) \\ &= f(0, X(0)) + \int_0^T f_t(t, X(t))dt + \int_0^T f_x(t, X(t))\Delta(t)dW(t) \\ &+ \int_0^T f_x(t, X(t))\Theta(t)dt + \frac{1}{2}\int_0^T f_{xx}(t, X(t))\Delta^2(t)d(t). \end{aligned} \tag{15}$$

Example: Geometric Brownian Motion

Let $W(t)$ be a BM, let $\mathcal{F}(t)$ be an associated filtration, and let $\alpha(t)$ and $\sigma(t)$ be adapted processes. Define the Itô process

$$X(t) = \int_0^t \sigma(t)dW(s) + \int_0^t (\alpha(s) - \frac{1}{2}\sigma^2(s))ds$$

Then

$$dX(t) = \sigma(t)dW(t) + (\alpha(t) - \frac{1}{2}\sigma^2(t))dt$$

and

$$dX(t)dX(t) = \sigma^2(t)dt$$

Example: Geometric Brownian Motion (Continued)

Consider an asset price process given by

$$S(t) = S(0)e^{X(t)} = S(0)\exp\int_0^t \sigma(t)dW(s) + \int_0^t (\alpha(s) - \frac{1}{2}\sigma^2(s))ds,$$

where $S(0)$ is nonrandom and positive. Let $S(t) = f(X(t))$, where $f(x) = S(0)e^x$. By Itô's formula,

$$\begin{aligned} dS(t) &= df(X(t)) \\ &= f'(X(t))dX(t) + \frac{1}{2}f''(X(t))d[X,X](t) \\ &= S(0)e^{X(t)}dX(t) + \frac{1}{2}S(0)e^{X(t)}\sigma^2(t)dt \qquad (16) \\ &= S(t)dX(t) + \frac{1}{2}S(t)\sigma^2(t)dt \\ &= \alpha(t)S(t)dt + \sigma(t)S(t)dW(t) \end{aligned}$$

It^o Integral of a Deterministic Integrand

Theorem 7

Let $W(s)$ be a Brownian motion, and let $\Delta(s)$ be a nonrandom function of time. Define $I(t) = \int_0^t \Delta(s)dW(s)$. For each $t \geq 0$, the random variable $I(t)$ is normally distributed with expected value zero and variance $\int_0^t \Delta^2(s)ds$.

Proof Since $I(t)$ is a martingale and $I(0) = 0$, we must have $\mathbb{E}I(t) = 0$.
By Itô's isometry, $VarI(t) = \mathbb{E}I^2(t) = \int_0^t \Delta^2(s)ds$.
To show that $I(t)$ is normally distributed, we use moment-generating function and martingale property.

Example: Vasicek Interest Rate Model

Let $R(t)$ be the interest rate process.

$$dR(t) = (\alpha - \beta R(t))dt + \sigma dW(t),\ \alpha, \beta, \sigma > 0.$$

Then

$$R(t) = e^{-\beta t}R(0) + \frac{\alpha}{\beta}(1 - e^{-\beta t}) + \sigma e^{-\beta t}\int_0^t e^{\beta s}dW(s)$$

To show this, we use the Itô formula with

$$f(t, x) = e^{-\beta t}R(0) + \frac{\alpha}{\beta}(1 - e^{-\beta t}) + \sigma e^{-\beta t}x$$

and $X(t) = \int_0^t e^{\beta s}dW(s)$.

Example: CIR Interest Rate Model

The CIR model for the interest raote process $R(t)$ is

$$dR(t) = (\alpha - \beta R(t))dt + \sigma\sqrt{R(t)}dW(t),\ \alpha, \beta, \sigma > 0.$$

The CIR equation does not have a closed-form solution. We instead derive the expected value and variance of $R(t)$. To do this, we use the function $f(t, x) = e^{\beta t}x$ and the Itô formula to compute

$$d(e^{\beta t}R(t)) = \alpha e^{\beta t}dt + \sigma e^{\beta t}\sqrt{R(t)}dW(t).$$

Integration of both sides yields

$$e^{\beta t}R(t) = R(0) + \frac{\alpha}{\beta}(e^{\beta t} - 1) + \sigma\int_0^t e^{\beta u}\sqrt{R(u)}dW(u).$$

Then

$$\mathbb{E}R(t) = e^{-\beta t}R(0) + \frac{\alpha}{\beta}(1 - e^{-\beta t}).$$

Example: CIR Interest Rate Model (Continued)

To compute the variance of $R(t)$, we set $X(t) = e^{\beta t} R(t)$. According to the Itô formula with $f(x) = x^2$,

$$d(X^2(t)) = 2\alpha e^{\beta t} X(t) dt + 2\sigma e^{\frac{\beta t}{2}} X^{\frac{3}{2}}(t) dW(t) + \sigma^2 e^{\beta t} X(t) dt.$$

Taking integration of both sides and then taking expectations, we can obtain $\mathbb{E}R^2(t)$. Finally,

$$Var(R(t)) = \frac{\alpha^2}{\beta} R(0)(e^{-\beta t} - e^{-2\beta t}) + \frac{\alpha\sigma^2}{2\beta^2}(1 - 2e^{\beta t} + e^{-2\beta t}).$$

In particular,

$$\lim_{t\to\infty} Var(R(t)) = \frac{\alpha\sigma^2}{2\beta^2}.$$

Black-Scholes-Merton Equation

Evolution of Portfolio Value

$$dS(t) = \alpha S(t)dt + \sigma S(t)dW(t)$$

Multiple Brownian Motions

Definition 3

A d-dimensional Brownian motion is a process

$$W(t) = (W_1(t), W_2(t), \cdots, W_d(t))$$

with the following properties.

(1) *Each $W_i(t)$ is a one-dimensional Brownian motion.*

(2) *If $i \neq j$, then the process $W_i(t)$ and $W_j(t)$ are independent.*

Two-dimensional Itô Formula

Theorem 8

Let $f(t,x,y) \in C^{1,2,2}$, let $X(t)$ and $Y(t)$ be Itô processes. Then

$$\begin{aligned} &df(t,X(t),Y(t)) \\ &= f_t(t,X(t),Y(t))dt + f_x(t,X(t),Y(t))dX(t) + f_y(t,X(t),Y(t))dY(t) \\ &+ \frac{1}{2}f_{xx}(t,X(t),Y(t))dX(t)dX(t) + f_{xy}(t,X(t),Y(t))dX(t)dY(t) \\ &+ \frac{1}{2}f_{yy}(t,X(t),Y(t))dY(t)dY(t). \end{aligned} \tag{17}$$

Itô Product Rule

Corollary: Itô Product Rule

Let $X(t), Y(t)$ be Itô processes. Then

$$d(X(t)Y(t)) = X(t)dY(t) + Y(t)dX(t) + dX(t)dY(t).$$

Recognizing a Brownian Motion

Theorem 9

(Levy, One Dimensional) Let $M(t), t \geq 0$ be a continuous martingale. Assume that $M(0) = 0$, and $[M, M](t) = t$ for all $t \geq 0$. Then $M(t)$ is a Brownian motion.

Theorem 10

(Levy, two dimensions) Let $M_1(t)$ and $M_2(t), t \geq 0$, be continuous martingales. Assume that $M_1(0) = M_2(0) = 0$, and $[M_i, M_j](t) = t\delta_{ij}$ for all $t \geq 0$. Then $M_1(t)$ and $M_2(t)$ are independent Brownian motions.

Example: Correlated Stock Prices

Suppose

$$\begin{aligned} dS_1(t) &= \alpha_1 S_1(t)dt + \sigma_1 S_1(t)dW_1(t), \\ dS_2(t) &= \alpha_2 S_2(t)dt + \sigma_2 S_2(t)\left[\rho dW_1(t) + \sqrt{1-\rho^2}dW_2(t)\right], \end{aligned} \tag{18}$$

where $W_1(t)$ and $W_2(t)$ are independent Brownian motions and $\sigma_1 > 0, \sigma_2 > 0$ and $-1 \leq \rho \leq 1$ are constant. Define

$$W_3(t) = \rho dW_1(t) + \sqrt{1-\rho^2}dW_2(t).$$

Then $W_3(t)$ is a continuous martigale with $W_3(0) = 0$, and

$$dW_3(t)dW_3(t) = dt.$$

Example: Correlated Stock Prices (Continued)

In other words, $[W_3, W-3](t) = t$. Thus, $W_3(t)$ is a Brownian motion. The Brownian motions $W_1(t)$ and $W_3(t)$ are correlated. Actually,

$$\mathbb{E}[W_1(t)W_3(t)] = \rho t$$

Because both $W_1(t)$ and $W_3(t)$ have standard deviation $\sqrt{t}$,

$$Corr(W_1(t), W_3(t)) = \rho$$